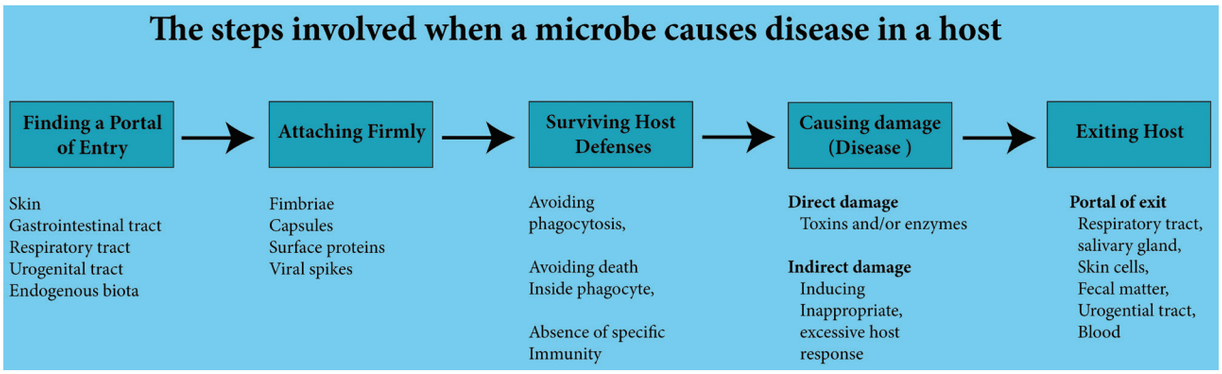


Figure 12.2: The steps involved when a microbe causes disease in a host

(shigellosis) from feces of infected people to food. Such vectors are called mechanical vectors.

Biological transmission is an active process and is more complex. The pathogens multiply in the body of the vectors often undergoing part of a developmental cycle in it. Such vectors are termed biological vectors. Example: *Aedes aegypti* mosquito transmitting dengue, *Anopheles* mosquito transmitting malaria.

Soil: Some pathogens are able to survive in the soil for very long periods. Spores of tetatus bacilli may remain viable in the soil for several decades and serve as the source of infection.

Water: Water may act as the source of infection due to contamination with pathogenic microorganisms. Example: Cholera causing *Vibrio cholerae*.

Food: Contaminated food materials may act as source of infection. The presence of pathogens in food may be due to external contamination. Example: Food contaminated by *Staphylococcus*.

12.1.3 Types of Infectious Diseases

Infectious diseases may be localised or generalised.

Localised infections: An infection that is restricted to a specific location or region within the body of the host is called localised infection.

Generalised infections: An infection that has spread to several regions or areas in the body of the host. This involves the spread of the infecting agent from the site of entry through tissue spaces or channels, along the lymphatics or through the bloodstream.

Circulation of bacteria in the blood is known as **Bacteremia**. **Septicemia** is the condition where bacteria circulate and multiply in the blood, form toxic products and cause high fever. **Pyemia** is a condition where pyogenic bacteria produce septicemia with multiple abscesses in the internal organs such as the spleen, liver and kidney.

Occurrence of a disease

To understand the full scope of a disease, we should know about its occurrence. Epidemiology involves in the study of the frequency and distribution of disease and other health related factors in defined populations. The incidence of a disease is the number of people in a population who develop a disease during a particular time period. The prevalence of a disease

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